

Has anyone estimated the amount of energy use of human brains compared to AI?

Posted by u/Raychao • • 0 points • 11 comments

I was watching a video last night by an English Teacher regarding AI. One of the arguments placed by this English Teacher was that AI was very expensive as an English writing tool because of its very high energy use. This got me thinking, which is the higher user of energy? The AI or the human brain?

My thoughts would be that the very long and uninterrupted chain of human brains have consumed enormous amounts of energy to write all the things human brains have written over history. Let's not even get started on the amount of energy that human brains have expended trying to get busy with other human brains to make yet more human brains to write more things and so on.

Has anyone compared the energy use of human brains versus AIs? Which base technology comes out in front on energy use to write a 5000 word essay?

Comments

u/[deleted] • 1 points • 2023-05-19 07:15 UTC

40 Wats

u/[deleted] • 5 points • 2023-05-18 07:47 UTC

Human brain: 1 exaflop

Top supercomputer: 1 exaflop

Currently we are equal. In theory.

But then again i dont really understand the researcher's analysis ... how on earth can a human performan 1 billion billion calculations per second???

But oh well...

Whatever it is ... it seems computers have surpassed us as they are only going to get higher than 1.1 quintillion calculations per second.

u/Rise-O-Matic • 10 points • 2023-05-17 23:38 UTC

Many smart professors suffer from the belief that they are a brain, and there's this body attached to it that's handy for moving them around.

Sure, a brain, if put in a jar and left to write, might be more efficient. But a human will use waaay more power in a day of writing, not from the writing itself, but from the consumption of just living their life while they write. And God forbid they have to *drive* somewhere to get paid to do it.

For the record, ChatGPT states that due to efficiencies gained via parallel processing, your essay prompt consumes a fraction of a joule, or less than a watt, of power to fulfill, though take that with a boulder of salt.

u/HalfSecondWoe • 7 points • 2023-05-17 23:10 UTC

In terms of raw energy use, brains are way more efficient. Food was scarce during most of our evolution, we're actually really good at starving without dying. The brain still uses up a huge chunk of that energy, but it's minimized for what it does

Processors use harder to find raw materials and push much more energy through, because they're optimized for performance, with energy consumption as a secondary concern. Electricity has been cheap from the day they were invented, it's just not what their engineering focuses on

If you want to add a bit of nuance to it, you could factor in the energy costs from farming compared to the energy costs from electricity generation. That would probably make it a little bit of a closer race

u/[deleted] • 1 points • 2023-05-17 22:04 UTC

[deleted]

u/xott • 8 points • 2023-05-17 21:49 UTC

Definitely the brain is more power efficient.

Generally runs somewhere about 20W and requires very little in infrastructure.

u/KidKilobyte • 16 points • 2023-05-17 22:05 UTC

But also takes more time, sometimes ridiculously more time, so you have to factor that in as watts per word produced

u/Surur • 10 points • 2023-05-17 22:59 UTC

I agree with you. It may take someone a day to write an essay and chatgpt 30 seconds. $20w \times 8 \text{ hrs} = 160wh$. That is say 5c of electricity. Say ChatGPT uses \$500,000 worth of electricity per day - that is 10 million essays per day. I think chatGPT can definitely do that and more, making it more energy efficient in terms of what per essay.

Now of course you can't switch off brains, but if you need 10 million essays, it would be better to use chatgpt to generate them than to create humans.

u/mast4pimp • 3 points • 2023-05-18 23:28 UTC

Brain does million other things with those 20watts, failed comparison.

u/Surur • 2 points • 2023-05-18 23:41 UTC

Acknowledged, but your boss does not really want to pay for you breathing.

u/Third_Party_Opinion • 5 points • 2023-05-17 22:48 UTC

It's pretty cool it can run on like, a banana. Banana in, 5000 word essay out.

u/[deleted] • 3 points • 2023-05-18 07:50 UTC

But thousands of bananas in order to sustain yourself to learn over the years until you are able to eat that nana and write that essay.